UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

| | |
|---|---|
| COURSE | PRE CALCULUS |
| COURSE CODE | MAT133 |
| EXAMINATION | OCTOBER 2026 |
| TIME | 2 HOURS |

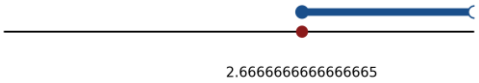
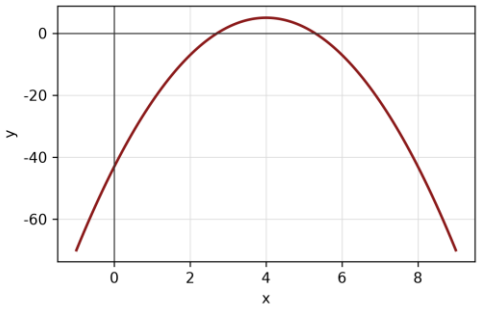
INSTRUCTIONS TO MARKERS

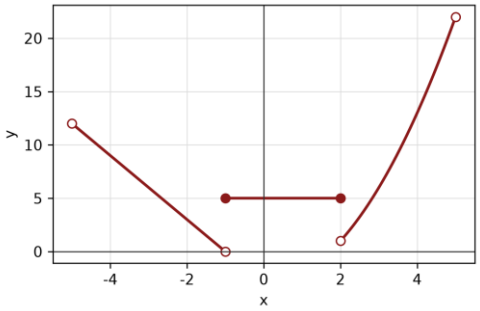
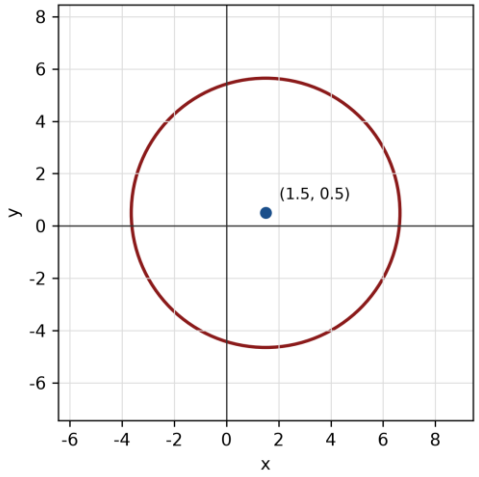
1. This marking scheme accompanies the MAT133 mock paper (October 2026).
2. Answers and marking points are for local study use only.

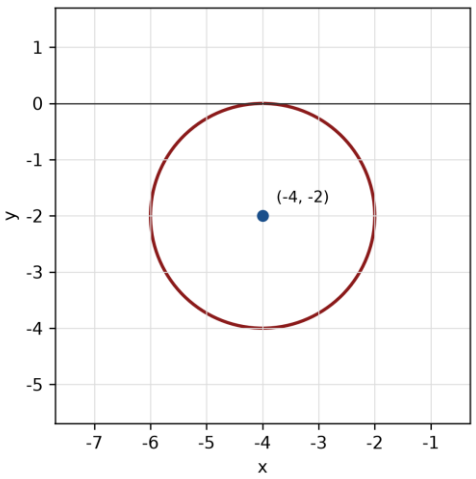
FOR MARKERS' USE ONLY

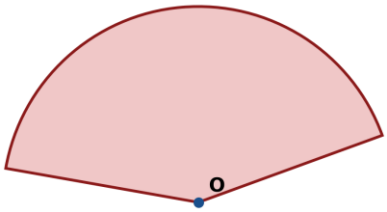
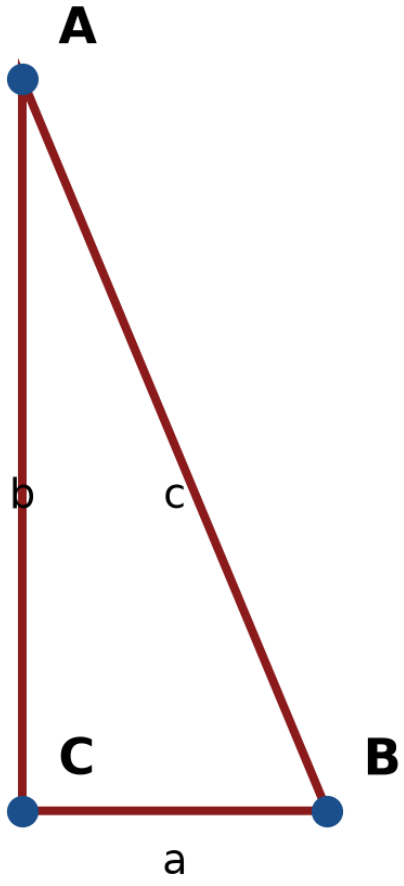
*This marking scheme consists of 5 printed pages*

«:> Hak Cipta Universiti Teknologi MARA

| NO | SOLUTION | | MARK |
|---|---|---|---|
| 1 a) | $5 - 3x \leq -3$<br>$-3x \leq -8$<br>$\left[\frac{8}{3}, \infty\right)$<br>  <br> | M1<br><br>A1 | 4 m |
| 1 b) | $\|3x + 3\| - 4 > 0$<br>$3x + 3 > 4$ or $3x + 3 \leq -4$<br>$\left(-\infty, -\frac{7}{3}\right) \cup \left(\frac{1}{3}, \infty\right)$<br>  | M1<br>A1<br>A1 | 5 m |
| 1 c) | $\|5x + 3\| = 8$<br>$5x + 3 = 8$<br>$5x + 3 = -8$ | M1<br>A1<br>A1 | 4 m |
| 2 a) | $y = 5 - 3(x - 4)^2$<br>vertex $(4, 5)$<br>$y(0) = -43$<br>  <br> | B1 B1<br><br>A1<br>A1 | 5 m |

| NO | SOLUTION | | MARK |
|---|---|---|---|
| 2 b) | $f(x) = -3x - 3, \quad x < -1$<br>$f(x) = 5, \quad -1 \le x \le 2$<br>$f(x) = x^2 - 3, \quad x > 2$ | B1 | 5 m |
| | $f(-2) = 3, \quad f(0) = 5, \quad f(3) = 6$ | A1 | |
| |  | | |
| 2 c) | $\left(\frac{3}{2}, \frac{1}{2}\right)$ | M1 | 5 m |
| | $r^2 = \frac{53}{2}$ | M1 | |
| | $\left(x - \frac{3}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{53}{2}$ | A1 | |
| |  | | |
| 3 a) | $\frac{1 - 4i}{-2 - i}$ | M1 | 5 m |
| | $= \frac{(1 - 4i)(-2 + i)}{(-2 - i)(-2 + i)}$ | M1 | |
| | $= \frac{2 + 9i}{5}$ | A1 | |
| | $\frac{2}{5} + \frac{9i}{5}$ | A1 | |

| NO | SOLUTION | | MARK |
|---|---|---|---|
| 3 b) | $\lvert z\rvert = \sqrt{-8^2 + 15^2} = 17$<br>$\tan\theta = \frac{15}{-8}, \quad \theta = 118.1^\circ$ | M1 A1 M1<br>A1 | 5 m |
| 3 c) | $-3x^3 + x^2 - 2x + 4 = (x+4)(-3x^2 + 13x - 54) + 220$<br>$q(x) = -3x^2 + 13x - 54, \quad r = 220$ | M1 M1 A1<br>A1 | 5 m |
| 4 a) | $-2x - 2y - 3z = 0$<br>$-x + 2y + 4z = -12$<br>$3x - 2y + z = 20$<br>$x = 4, \quad y = -4, \quad z = 0$ | M1<br>A1<br><br>A1 | 5 m |
| 4 b) | $-5x + y = 8$<br>$4x - 5y = 2$<br>$x = -2, \quad y = -2$ | M1<br>A1<br>A1 | 4 m |
| 4 c) | $(x+4)^2 + (y+2)^2 = 4$<br>$x^2 + 8x + y^2 + 4y + 20 = 4$<br>$x^2 + 8x + y^2 + 4y + 16 = 0$<br>  | M1<br>A1<br>B1 | 5 m |
| 5 a) | $\sin^2(x) - 1 = 0$<br>$(\sin(x) - 1)(\sin(x) + 1) = 0$<br>$\sin x = -1$ or $\sin x = 1$<br>$x = 90^\circ, 270^\circ$ | M1<br>A1<br><br>A1 | 5 m |

| NO | SOLUTION | MARK |
|---|---|---|
| 5 b) | $\theta = \frac{5\pi}{6}$ M1 M1<br>$A = \frac{1}{2}r^2\theta = \frac{1}{2}(6)^2\left(\frac{5\pi}{6}\right) = 15\pi$ A1<br>  | 4 m |
| 5 c) | $5^2 + 12^2 = 13^2$ M1<br>$\tan A = \frac{5}{12} \rightarrow A = 22.62^\circ$ A1<br>$B = 90^\circ - A = 67.38^\circ$ A1<br>  | 5 m |